

AV NETWORK DESIGN REPORT

# V12 Many 10G closets, core port pressure

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**NOT VALIDATED**

Estimate for planning only. Send this design to the Pro AV Design team for validation before ordering.

**36**

DEVICES

**8**

SWITCHES

**0 W**

POE LOAD

**1.7 kW**

EST. POWER

**8.0 U**

RACK SPACE

# Networking engineered for AV over IP

NETGEAR AV switches, routers and WiFi access points are built specifically for networked media and AV, with out-of-the-box performance and easy setup for even the most complex installation. NETGEAR works with AV manufacturers and integration partners around the world, and supports the partners and associations that make the Pro AV and broadcast industries work.

## ● Optimized for AV over IP

Low-latency switching for audio, video and control, with PoE+ (30 W) and Ultra PoE++ (90 W) per port options.

## ● Certified AV profiles

NETGEAR AV OS with configuration profiles certified by leading AV manufacturers (Audinate, Crestron, Q-SYS, SMPTE ST 2110, NDI).

## ● Multicast that just works

NETGEAR IGMP Plus stops multicast flooding automatically, and Auto-LAG and Auto-Trunk connect switches without manual setup.

## ● Free central management

The NETGEAR Engage Controller manages switches, routers and WiFi 7 access points from one screen, and can set up the network before the hardware arrives.

## ● Scales with the project

From a single rack to redundant, active-active cores with 10G, 25G and 100G uplinks.

## ● Lower total cost

Energy-efficient hardware, a lifetime warranty on managed switches and less need for specialist IT resources.

### IN THIS DESIGN

This design uses M4350 and M4500 switches, all managed for free from one screen in the NETGEAR Engage Controller with NETGEAR AV OS.

### BUILT FOR EVERY SPACE

Broadcast and live events · Corporate and commercial AV · Higher education · Medical · Unified communications · High-end residential

### HELP FROM DESIGN TO DEPLOYMENT

NETGEAR has a dedicated Pro AV and broadcast network design team, pre-sales support and technical assistance. Free on-demand training and certification at [academy.netgear.com](https://academy.netgear.com) covers networking basics through in-depth troubleshooting, and the top courses are AVIXA-accredited for CTS renewal units.

# Support and services

Every NETGEAR managed switch includes a lifetime hardware warranty and complimentary support to get you started. Premium Support tiers add guaranteed response times, longer support hours and faster hardware replacement, so you can match support to how critical the system is.

|                                    | Warranty (included) | Sprint              | Overdrive         | Fastlane |
|------------------------------------|---------------------|---------------------|-------------------|----------|
| Complimentary setup support        | ✓                   | ✓                   | ✓                 | ✓        |
| AI-driven chat support             | ✓                   | ✓                   | ✓                 | ✓        |
| Direct access to support engineers |                     | ✓                   | ✓                 | ✓        |
| Support hours                      |                     | 24/5                | 24/7              | 24/7     |
| Guaranteed response time (P1)      |                     | 8 hours             | 2 hours           | Custom   |
| Priority case handling             |                     |                     | ✓                 | ✓        |
| Escalation to senior engineers     |                     |                     | Level 2           | Level 3  |
| Hardware replacement               | Fast RMA            | Advance replacement | Next business day | Custom   |
| Scheduled support calls            |                     |                     | ✓                 | ✓        |
| Operational reports                |                     |                     | Quarterly         | Custom   |
| Designated or dedicated engineer   |                     |                     |                   | ✓        |
| Custom workflows and SLAs          |                     |                     |                   | ✓        |

Managed and enterprise products include 90 days of complimentary phone, chat and online support. Fully managed switches include 3 years of Sprint Support. Sprint and Overdrive response times are engagement commitments for Priority 1 cases.

## PROFESSIONAL SERVICES

|                                         |                                                                                                                                             |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Deployment services                     | Design validation, hands-on configuration and readiness testing, so the network is production-ready from day one. Remote, onsite or hybrid. |
| Health checks and optimization          | Proactive network audits that find issues before they become outages, with a clear remediation plan.                                        |
| Custom training and enablement          | Training based on your deployed system: architecture walkthroughs, feature deep dives, best practices and troubleshooting.                  |
| Interoperability, staging and migration | Multi-vendor compatibility checks, staged configurations and low-risk migrations to NETGEAR.                                                |
| Event services                          | Pre-event validation and real-time onsite or remote engineering support for live broadcasts, sporting events and other high-stakes shows.   |
| Fastlane professional engagement        | A high-touch engagement with senior engineering access for organizations that need deeper collaboration.                                    |
| Part number                             | Service                                                                                                                                     |
| PSP1104-10000S                          | On-site engineering assistance, full day                                                                                                    |
| PRC0001-10000S                          | New remote configuration, full day                                                                                                          |

Pricing depends on project scope and the level of support, and includes travel and expenses. Ask your NETGEAR contact or distributor for a quote.

# Design summary

Estimate for planning only. This bill of materials is generated automatically as a budgetary estimate. It must be validated by the ProAV Design team before ordering. Validation: [proavdesign@netgear.com](mailto:proavdesign@netgear.com)

## ROOMS AND DEVICES

| Device                                            | Category | Qty | Link       | PoE each |
|---------------------------------------------------|----------|-----|------------|----------|
| Main equipment room • MDF • 0 devices             |          |     |            |          |
| Closet 1 • IDF, 100 m multimode fiber • 6 devices |          |     |            |          |
| 10G AV-over-IP encoder (copper)                   | Video    | 6   | 10G Copper | –        |
| Closet 2 • IDF, 100 m multimode fiber • 6 devices |          |     |            |          |
| 10G AV-over-IP encoder (copper)                   | Video    | 6   | 10G Copper | –        |
| Closet 3 • IDF, 100 m multimode fiber • 6 devices |          |     |            |          |
| 10G AV-over-IP encoder (copper)                   | Video    | 6   | 10G Copper | –        |
| Closet 4 • IDF, 100 m multimode fiber • 6 devices |          |     |            |          |
| 10G AV-over-IP encoder (copper)                   | Video    | 6   | 10G Copper | –        |
| Closet 5 • IDF, 100 m multimode fiber • 6 devices |          |     |            |          |
| 10G AV-over-IP encoder (copper)                   | Video    | 6   | 10G Copper | –        |
| Closet 6 • IDF, 100 m multimode fiber • 6 devices |          |     |            |          |
| 10G AV-over-IP encoder (copper)                   | Video    | 6   | 10G Copper | –        |

# Network diagram



Green lines are uplinks, labelled with link count, speed and the optic or cable used. With a redundant core, every switch has its own full-capacity link group to each core, so either core can carry all traffic alone.

## Bill of materials

| Part number            | Description                         | Qty       | Where                                                                                  |
|------------------------|-------------------------------------|-----------|----------------------------------------------------------------------------------------|
| <b>SWITCHES</b>        |                                     |           |                                                                                        |
| <b>XSM4340V-100NES</b> | M4350-24X8F8V switch                | <b>6</b>  | Closet 1, Closet 2, Closet 3, Closet 4, Closet 5, Closet 6                             |
| <b>XSM4556-100NAS</b>  | M4500-48XF8C core switch            | <b>2</b>  | Main equipment room (core)                                                             |
| <b>OPTICS</b>          |                                     |           |                                                                                        |
| <b>NGS25G-SR-OC</b>    | 25GBASE-SR SFP28, multimode, 850 nm | <b>96</b> | Closet 1, Main equipment room (core), Closet 2, Closet 3, Closet 4, Closet 5, Closet 6 |
| <b>CABLING</b>         |                                     |           |                                                                                        |
| <b>ACC761-10000S</b>   | 100G QSFP28 DAC 1m, passive copper  | <b>2</b>  | Main equipment room (core)                                                             |

Part numbers are for Americas. Pricing and availability are provided by NETGEAR or your distributor after validation.

## Power and rack space

| Room                              | Est. draw | Datasheet max | Heat load    | Current        | Circuits | UPS size | Rack  |
|-----------------------------------|-----------|---------------|--------------|----------------|----------|----------|-------|
| <b>Closet 1</b>                   | 152 W     | 2,057 W       | 518 BTU/hr   | 1.4 A at 110 V | 1 × 20 A | 169 VA+  | 1.0 U |
| <b>Closet 2</b>                   | 152 W     | 2,057 W       | 518 BTU/hr   | 1.4 A at 110 V | 1 × 20 A | 169 VA+  | 1.0 U |
| <b>Closet 3</b>                   | 152 W     | 2,057 W       | 518 BTU/hr   | 1.4 A at 110 V | 1 × 20 A | 169 VA+  | 1.0 U |
| <b>Closet 4</b>                   | 152 W     | 2,057 W       | 518 BTU/hr   | 1.4 A at 110 V | 1 × 20 A | 169 VA+  | 1.0 U |
| <b>Closet 5</b>                   | 152 W     | 2,057 W       | 518 BTU/hr   | 1.4 A at 110 V | 1 × 20 A | 169 VA+  | 1.0 U |
| <b>Closet 6</b>                   | 152 W     | 2,057 W       | 518 BTU/hr   | 1.4 A at 110 V | 1 × 20 A | 169 VA+  | 1.0 U |
| <b>Main equipment room (core)</b> | 788 W     | 788 W         | 2,689 BTU/hr | 7.2 A at 110 V | 1 × 20 A | 876 VA+  | 2.0 U |

### POWER SUPPLIES PER SWITCH

| Room                       | Device        | Qty | PoE load each | Power supplies             | PoE available | Headroom | Est. draw each |
|----------------------------|---------------|-----|---------------|----------------------------|---------------|----------|----------------|
| Closet 1                   | M4350-24X8F8V | 1   | –             | Base (no module)           | 290 W         | 100%     | 152 W          |
| Closet 2                   | M4350-24X8F8V | 1   | –             | Base (no module)           | 290 W         | 100%     | 152 W          |
| Closet 3                   | M4350-24X8F8V | 1   | –             | Base (no module)           | 290 W         | 100%     | 152 W          |
| Closet 4                   | M4350-24X8F8V | 1   | –             | Base (no module)           | 290 W         | 100%     | 152 W          |
| Closet 5                   | M4350-24X8F8V | 1   | –             | Base (no module)           | 290 W         | 100%     | 152 W          |
| Closet 6                   | M4350-24X8F8V | 1   | –             | Base (no module)           | 290 W         | 100%     | 152 W          |
| Main equipment room (core) | M4500-48XF8C  | 2   | –             | 2x APS750W (1+1, included) | –             | –        | 394 W          |

Estimated draw = datasheet maximum without PoE + PoE load ÷ 0.9 (PSU efficiency). Circuits assume 80% continuous load.

# Checks

|      |                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NOTE | Faster uplinks (fewer links) give a smaller overall design than the slowest uplink speed, so they were used.                                                                                                            |
| NOTE | Inter-core link sized for half of all room traffic (active/active cores), and never below the busiest single switch: needs 180 Gbps, provided $2 \times 100G = 200$ Gbps.                                               |
| NOTE | Switches interconnect with LAG uplinks and IGMP Plus (no stacking), which keeps AVB and PTP available.                                                                                                                  |
| NOTE | Uplink LAGs use 1, 2, 4 or 8 links to each core, and each core link group carries the switch's full load on its own so one core can fail (sizes from Tool_Settings: 1, 2, 4, 8), so LAG hashing spreads traffic evenly. |

## UPLINK PLAN

| Room     | Switch        | Units | Uplinks each | Speed | Run        | Optic / cable |
|----------|---------------|-------|--------------|-------|------------|---------------|
| Closet 1 | M4350-24X8F8V | 1     | 8            | 25G   | MMF, 100 m | NGS25G-SR-OC  |
| Closet 2 | M4350-24X8F8V | 1     | 8            | 25G   | MMF, 100 m | NGS25G-SR-OC  |
| Closet 3 | M4350-24X8F8V | 1     | 8            | 25G   | MMF, 100 m | NGS25G-SR-OC  |
| Closet 4 | M4350-24X8F8V | 1     | 8            | 25G   | MMF, 100 m | NGS25G-SR-OC  |
| Closet 5 | M4350-24X8F8V | 1     | 8            | 25G   | MMF, 100 m | NGS25G-SR-OC  |
| Closet 6 | M4350-24X8F8V | 1     | 8            | 25G   | MMF, 100 m | NGS25G-SR-OC  |